

18-07-2026

Agenda: ML-20

Team will upload
wednesday still recording
soon. sorry for the day

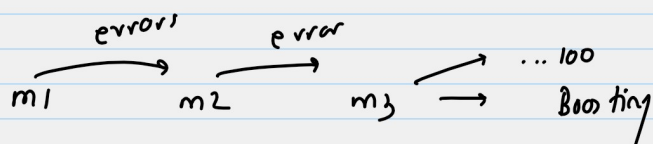
Gradient Boosting } Regression •
Xgboost
practicals • + docker intro

please read this

Gradient Boosting & xgboost does not use mean, no
bootstrap data.

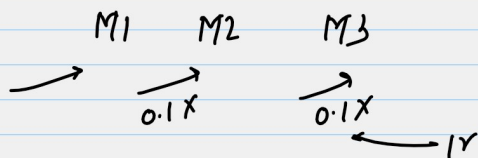
Data → 0.8 subsample → 80% of the dataset

oob → 0.8 → train
→ 0.2 → oob data



Gradient Boosting

↓ technique
• sequential models



DT → simple models
→ stumps [1, 3, 6, 8]

$$\text{loss function} = \frac{1}{2} (y - \hat{y})^2$$

Dataset

ID	Size	Beds	price (y)
1	800	2	120
2	900	2	130
3	1000	3	150
4	1100	3	170
5	1200	4	200

Step: 0 $\rightarrow$ Initial model (f_0)

$$f0 = \text{mean}(y) \rightarrow (120 + 130 + 150 + 170 + 200) / 5 = 154$$

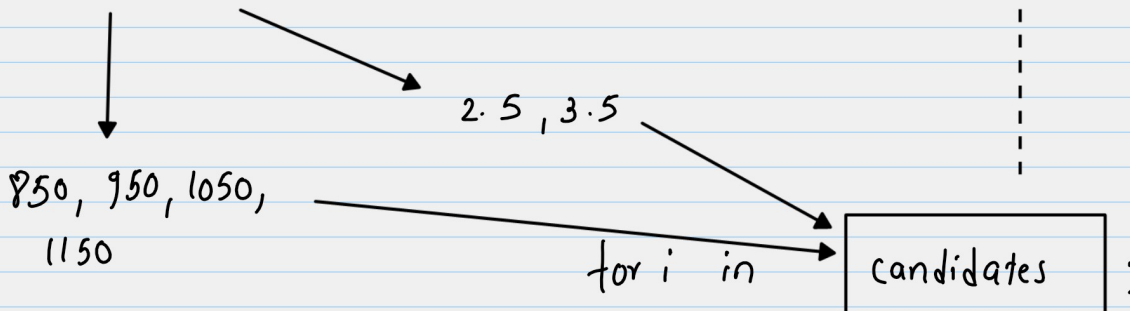
→ calculate residuals $(y - f) = (y - f_0)$

ID	Size	Beds	price (y)	$r_1(y - f_0)$
1	800	2	120	-34
2	900	2	130	-24
3	1000	3	150	-4
4	1100	3	170	+16
5	1200	4	200	+46

target for the stump

→ candidate splits

ID	Size	Beds	price (y)	$r_1(y - f_0)$
1	800	2	120	-34
2	900	2	130	-24
3	1000	3	150	-4
4	1100	3	170	+16
5	1200	4	200	+46



$(y - \text{mean})^2$

→	left (A)	right (B)	
→	mean (A)	mean (B)	← residual
→	SSE (residual) over both sides		

mean

$$|v| = 0.1$$

Data (X)

model [fo]

154

GB(Regression)

$$\text{loss} = \frac{1}{2} (y - f(x))^2$$

gradient (first derivative)

$$\rightarrow g = \frac{\partial L}{\partial f} = f - \gamma$$

changing direction to lower the loss $\rightarrow$

$$-g = \gamma - f$$

Feature	Threshold	left (A)	Right (B)	mean (A)	mean (B)	SSE
Size	1050	[1, 2, 3]	[4, 5]	-20.66	31	916 ← lowest
Size	950	[1, 2]	[3, 4, 5]	-29	19.33	1316.66
Size	1150	[1, 2, 3, 4]	[5]	-11.5	46	1475
Size	850	[1]	[2, 3, 4, 5]	-34	8.5	2675
Beds	2.5	[1, 2]	[3, 4, 5]	-29	19.33	1316.66
Beds	3.5	[1, 2, 3, 4]	[5]	-11.5	46	1475

Size < 1050 chosen as stump

left → right

full calculation example below

ID	size	residual	ID	size	residual
1	800	-34	4	1100	16
2	900	-24	5	1200	46
3	1000	-4			

mean = -20.66

mean = 31

← from above table

$SSE(A) =$

$SSE(B) =$

$$\begin{aligned}
 (-34 - (-20.66)) &= -13.33 \rightarrow \text{squared} = 177 \\
 (-24 - (-20.66)) &= -3.33 \rightarrow = 11.11 \\
 (-4 - (-20.66)) &= 16.66 \rightarrow = 277.77 \\
 \hline
 &466
 \end{aligned}$$

$$\begin{aligned}
 (16, 31) &= -15 \rightarrow \text{squared} = 225 \\
 (46, 31) &= 15 \rightarrow 225 \\
 \hline
 &450
 \end{aligned}$$

$$\begin{aligned}
 \text{Total SSE} &= SSE(A) + SSE(B) \\
 &= 916.66
 \end{aligned}$$

$f_0(x) \rightarrow 154$

← model-1

$h_1(x) \rightarrow$

 Size ≤ 1050

 ↙ ↘

 -20.66 31

← model-2

$$F_1(x) = f_0 + r * h_1(x)$$

ID	Size	Beds	price (y)	$r_1(y - f_0)$	f_1	$r_2(y - f_1)$
1	800	2	120	-34	151.93	-31.93
2	900	2	130	-24	151.93	-21.93
3	1000	3	150	-4	151.93	-1.93
4	1100	3	170	+16	157.1	12.90
5	1200	4	200	+46	157.1	42.90

compute f_1 :

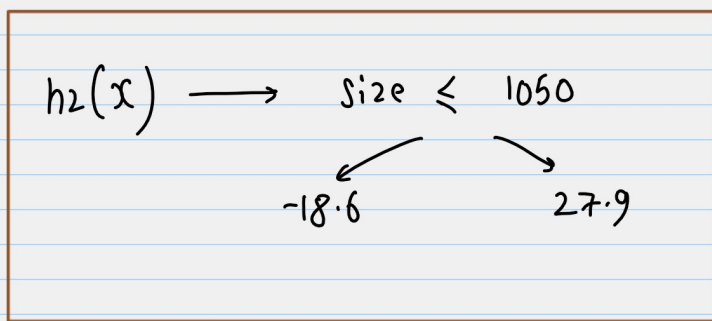
$$10-1(A) : 154 + 0.1 \times (-20.66) = 151.93$$

$\uparrow$
 $800 \leq 1050$
 $\swarrow \quad \searrow$
 $-20.66 \quad 31.0$

we will use r_2 as the next stump

Feature	Threshold	left(A)	right(B)	mean(A)	mean(B)	SSE
Size	1050	[1,2,3]	[4,5]			
Size	950	[1,2]	[3,4,5]			
Size	1150	[1,2,3,4]	[5]			
Size	850	[1]	[2,3,4,5]			
Beds	2.5	[1,2]	[3,4,5]			
Beds	3.5	[1,2,3,4]	[5]			

Feature	Threshold	left(A)	right(B)	mean(A)	mean(B)	SSE	
Size	1050	[1,2,3]	[4,5]	-18.60	27.90	916.66	← lowest
Size	950	[1,2]	[3,4,5]	-26.93	17.95	1093.35	
Size	1150	[1,2,3,4]	[5]	-10.72	42.9	1210.85	
Size	850	[1]	[2,3,4,5]	-31.93	8.5	2236.69	
Beds	2.5	[1,2]	[3,4,5]	-26.93	17.95	1093.35	
Beds	3.5	[1,2,3,4]	[5]	-10.72	42.9	1210.85	



→ new stump

→ model-3

$$f_1(x) = f_0 + \gamma * h_1(x)$$

$$f_2(x) = f_0 + \gamma * h_1(x) + \gamma * h_2(x)$$

$$= f_1(x) + \gamma * h_2(x)$$

↙ copied from above for f_1 value

ID	Size	Beds	price (y)	$r_1(y - f_0)$	f_1	$r_2(y - f_1)$
1	800	2	120	-34	151.93	-31.93
2	900	2	130	-24	151.93	-21.93
3	1000	3	150	-4	151.93	-1.93
4	1100	3	170	+16	157.1	+2.90
5	1200	4	200	+46	157.1	+2.90

ID	Size	Beds	price (y)	$r_1(y - f_0)$	f_1	$r_2(y - f_1)$	f_2	$r_3(y - f_2)$
1	800	2	120	-34	151.93	-31.93	150.07	-30.07
2	900	2	130	-24	151.93	-21.93	150.07	-20.07
3	1000	3	150	-4	151.93	-1.93	150.07	-0.07
4	1100	3	170	+16	157.1	+2.90	159.89	+10.11
5	1200	4	200	+46	157.1	+2.90	159.89	+40.11

compute - f_2

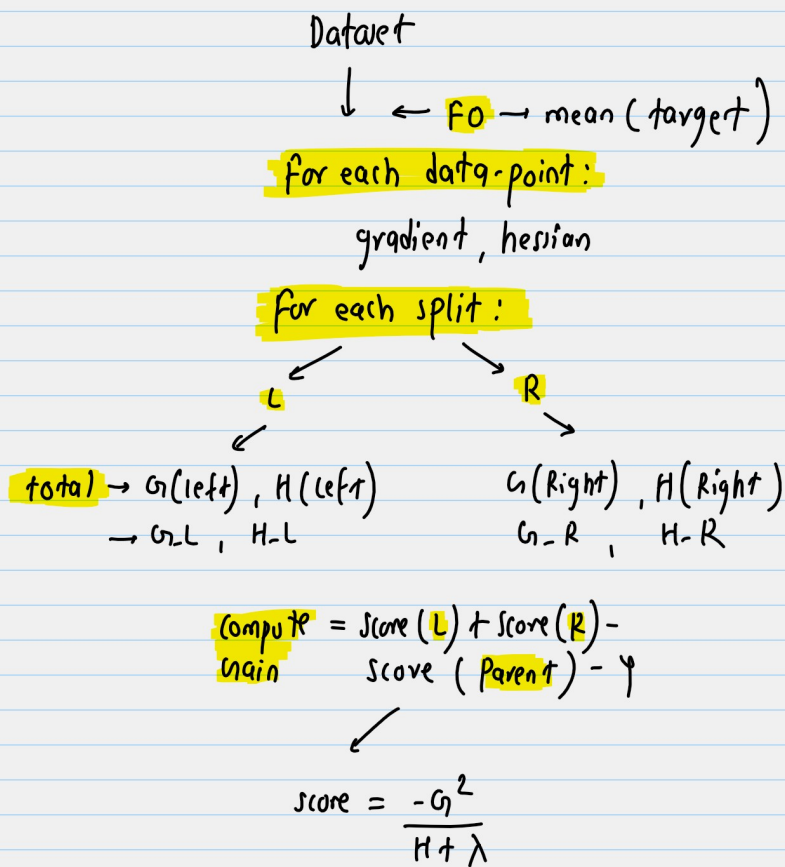
$$ID = f_1(x) + \gamma * h_2(x)$$

$$[1,2,3] = 151.93 + 0.1 * (-18.6) = 150.07$$

$$ID = 157.1 + 0.1 * 27.9 = 159.89$$

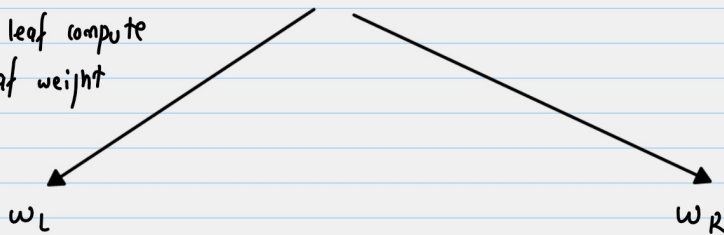
Because the residuals maintain the same sign pattern (IDs 1-3 negative, 4-5 positive), splitting along Size still best separates negative vs positive residuals and thus minimizes SSE. As iterations progress residual magnitudes shrink, and at some point another split (e.g., Beds or a different Size cutoff) might become best, but early on Size is the dominant predictive feature.

Xtreme - gradient boost



choose-split with highest possible gain

for each leaf compute final leaf weight



Final prediction $(f_{t+1}) = f_0 + \eta \cdot f_t(x)$

$g_B \leftarrow x_{gB}$

loss = squared error

$$l = \frac{1}{2} (y - \hat{y})^2$$

$$\text{gradient} = \frac{\partial L}{\partial \hat{y}} = \hat{y} - y$$

$$\text{hessian} = \frac{\partial^2 L}{\partial \hat{y}^2} \approx 1$$

$$\lambda(\text{regularization}) = 1$$

$$\varphi(\text{minimum gain to split}) = 0$$

$\lambda \rightarrow$ denominator (L2)
 $\rightarrow$ shrinks core

$\gamma \rightarrow$ subtraction penalty (11)

α $\rightarrow$ leaf weight calc,
not split core (1r)

$$\omega = \frac{-G}{H + \lambda}$$

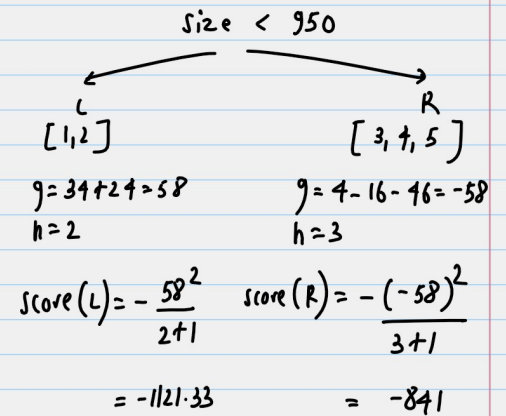
$\eta = 1$
for simple calculation

ID	Size	Beds	price (y)	$f_0(\hat{y}_i^{(0)})$	$g = \hat{y}_0 - y$	h
1	800	2	120	154	34	1
2	900	2	130	154	24	1
3	1000	3	150	154	4	1
4	1100	3	170	154	-16	1
5	1200	4	200	154	-46	1

Iteration-1 → Build Tree

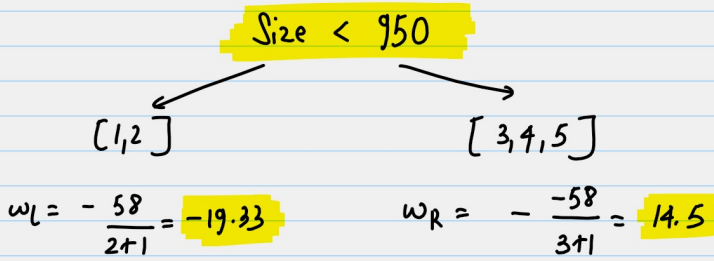
Size	850	gain
Size	950	1962.33
Size	1050	707
Beds	2.5	1156
Beds	3.5	707

highest gain → calculation



leaf-weights:

$$w = -\frac{g}{H + \lambda}$$



score(Parent)

$$g = 34 + 24 + 4 - 16 - 46 = 0$$

$$h = 5$$

$$= -\frac{0^2}{5+1} = 0$$

$$\text{Gain} = 1121.33 + 841 = 1962.33$$

$$(f_{t+1}) = f_0 + \eta \cdot f_t(x)$$

ID	Size	Beds	price (y)	$f_0(\hat{y}_i^{(0)})$	leaf	weight	F_i
1	800	2	120	154	L	-19.33	134.67
2	900	2	130	154	L	-19.33	134.67
3	1000	3	150	154	R	+14.5	168.5
4	1100	3	170	154	R	+14.5	168.5
5	1200	4	200	154	R	+14.5	168.5

$$10-1 \rightarrow 154 + 1 * (-19.33) = 134$$

Iteration-2

ID	Size	Beds	price (y)	F_i	g	h
1	800	2	120	134.67	14.67	1
2	900	2	130	134.67	4.67	1
3	1000	3	150	168.5	18.5	1
4	1100	3	170	168.5	-1.5	1
5	1200	4	200	168.5	-31.5	1

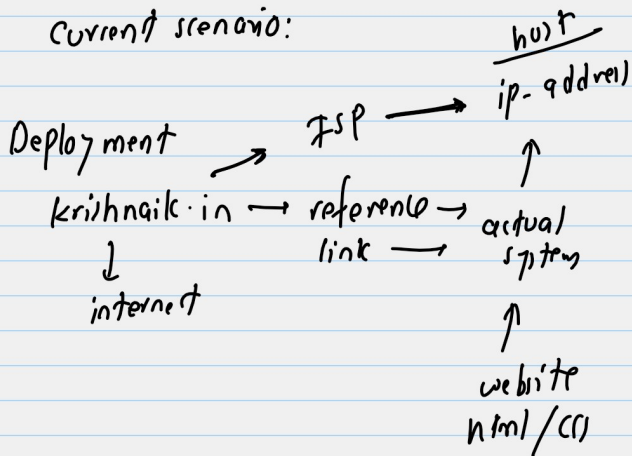
Gradient-Boost

Xgboost

Use derivative of loss	Yes	Yes
Use second order derivative	No	Yes
split score	SSE	Based on gain (g, h, λ)
Regularization	weak (α)	Strong $(\lambda, \alpha, \gamma)$
Tree complexity control	max-depth	prunes automatically ↑ max-depth
leaf value	mean residual	$-\frac{g}{H + \lambda}$
system speed	slower	much faster, parallel

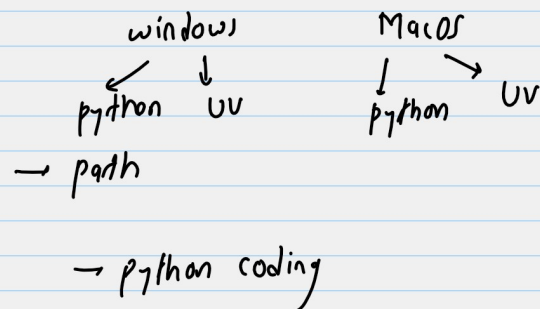
Docker:

current scenario:

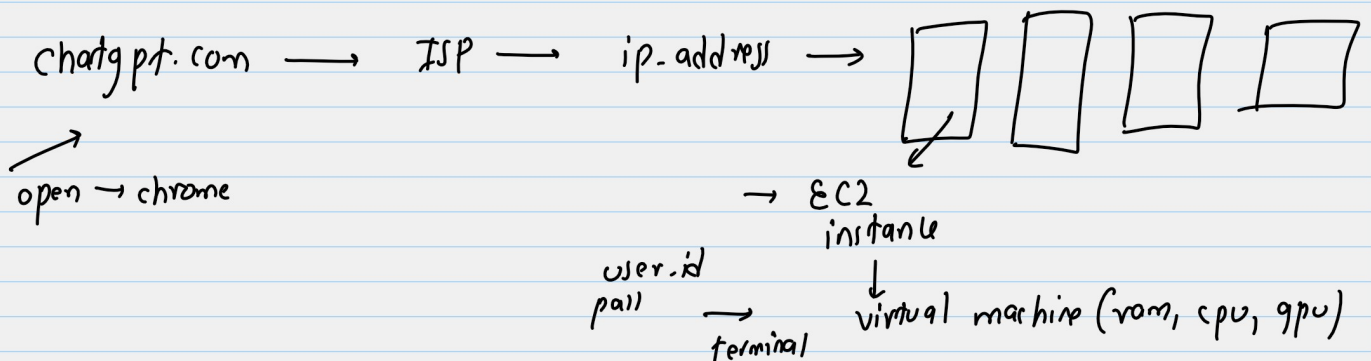


UDS 2.0

system setup (person system)

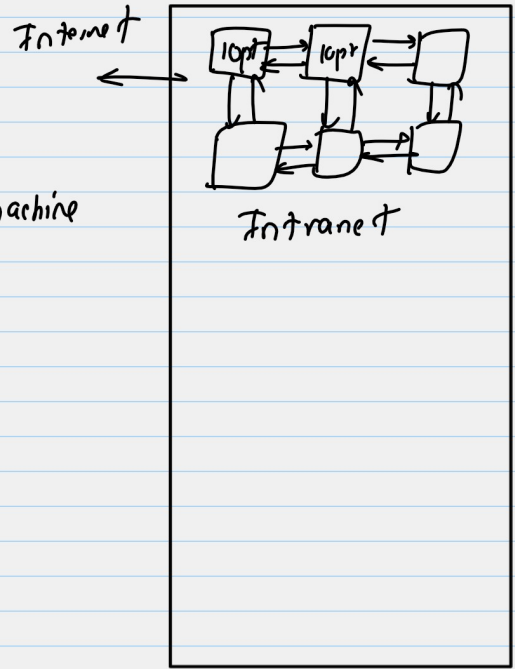


AWS GCP Azure cloudflare



CPU, cache d, computer — physical machine ← Any desk
 (2 core)
 (8gb ram)
 (1gb gpu)
 (256 gb store)

↑
 start
 login



for

virtual machine

id →
 password

EC2 → 1
 → 100gb ram
 → 100gb gpu
 → 200 core cpu
 → 50 prt store

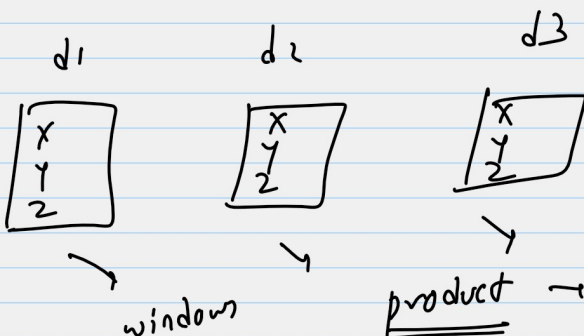
Amazon Data centre
 (USA)

Deployment → server → power system
 ↓
 or → stable → full control
 → light weight
 ↓
 linux → version
 flavour
 ← open source

linux doc
 ↓
 docker
 EC2
 instance
 → linux flavour

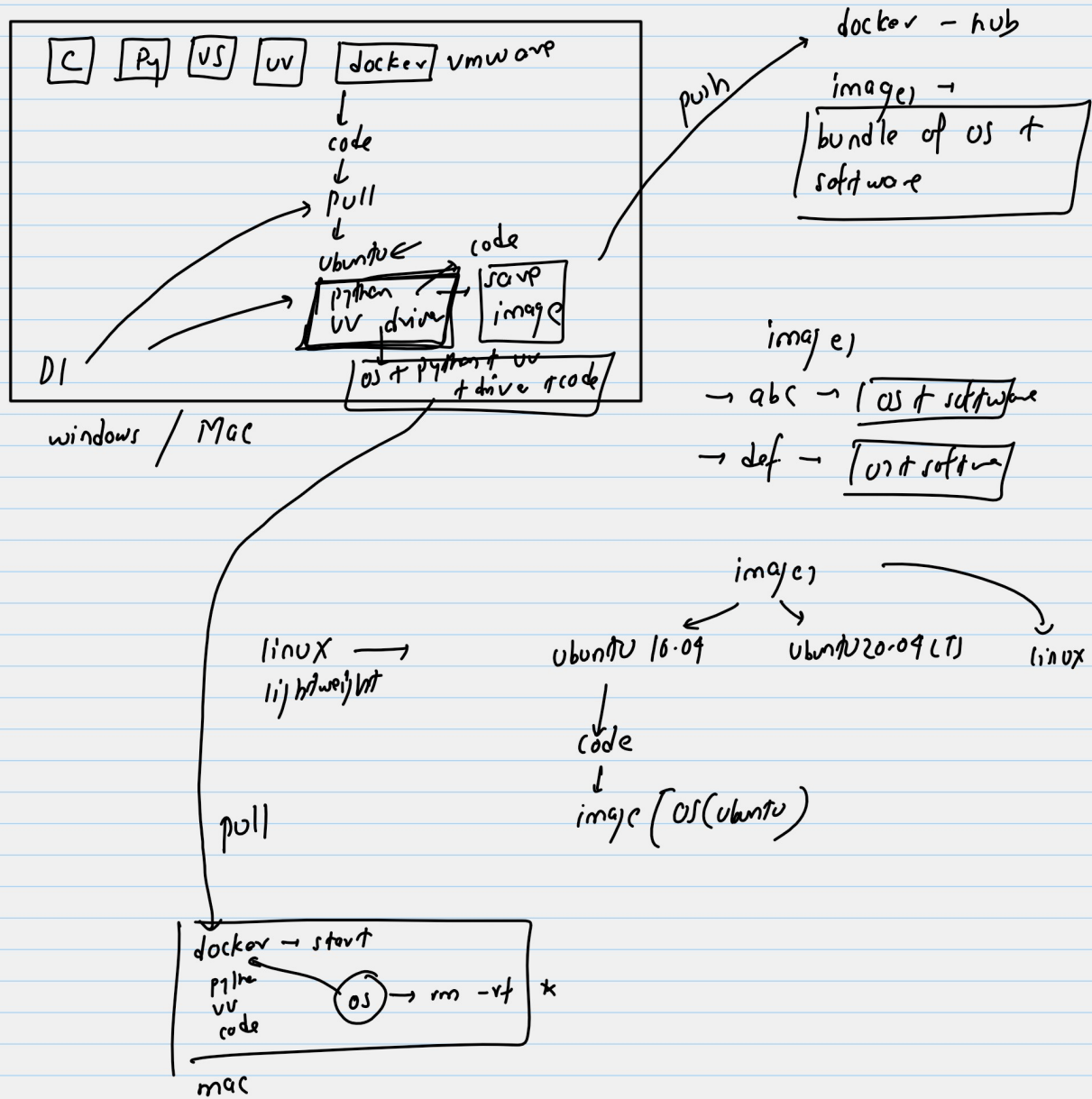
developer → windows → python 3.6 → computer vision
 → version of driver gpu
 → 11
 → mac → python 3.14 → ML
 → windows → t1
 → python 3.6
 → cpu & version

↓
compatibility
 ↓
same environment

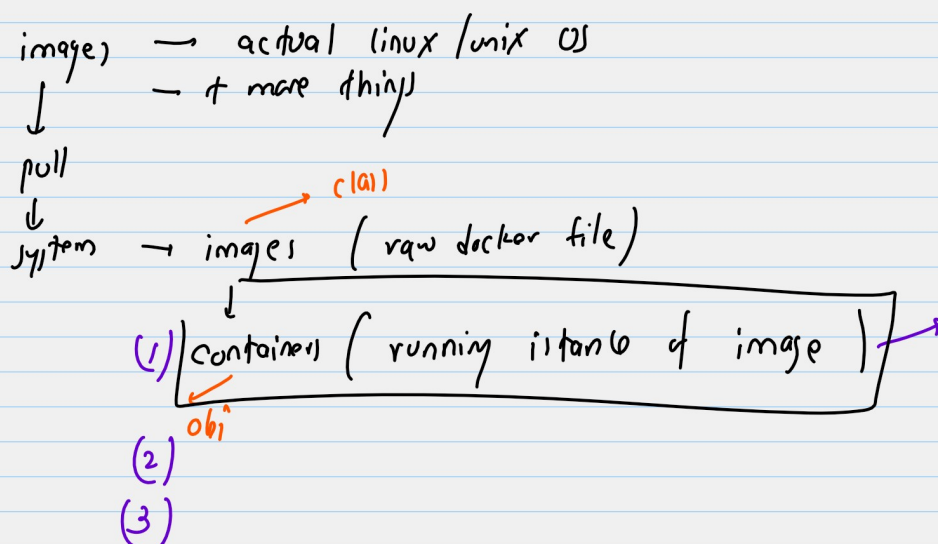


product → deployment → linux → EC2
 → x
 y
 z

- redundant effort
 - system setup
 - manual work
- docker → open source



D2



Docker Engine : The underlying client-server application that manages images, container, network & data volumes

- image
- container
- go inside
- install
- build local
- push

FROM python:3.12-slim-trixie

← ← lighter weight ^{docker-image}

The installer requires curl (and certificates) to download the release archive

RUN apt-get update && apt-get install -y --no-install-recommends curl ca-certificates

Download the latest installer

ADD https://releases.astral.sh/installers/uv/latest/uv-installer.sh

Run the installer then remove it

RUN sh /uv-installer.sh && rm /uv-installer.sh

Ensure the installed binary is on the `PATH`

ENV PATH="/root/.local/bin/:\$PATH"

